

Yashwanth Yarlagadda

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Summary

Software Development Engineer with 4+ years of experience designing and building scalable backend systems, cloud-native applications, and AI-powered services on AWS. Experienced in Java, Python, TypeScript, Spring Boot, AWS serverless technologies, Amazon SageMaker AI, and large language model (LLM) applications. Strong background in developing production-grade ML infrastructure, distributed systems, and automation solutions, with a focus on reliability, scalability, and operational excellence.

Experience

Amazon

Software Development Engineer

Seattle, WA

August 2022 - Present

Customer Service Wit Team

SmartMatch & Contact Summarizer Models

- Designed, deployed, and maintained Amazon SageMaker models and inference endpoints supporting SmartMatch and Contact Summarizer, enabling scalable production-grade ML inference.
- Designed and implemented FAST stack integration for SmartMatch, replacing the legacy model invocation path and improving inference reliability.
- Developed Python-based inference-layer components implementing request pre-processing, feature transformation, response post-processing, and seamless integration with SageMaker endpoints.
- Deployed a secondary SageMaker endpoint trained on redacted customer transcripts to satisfy Amazon Security Review (ASR) requirements for sensitive customer data.
- Built CloudWatch p99 latency metrics, enabling proactive monitoring and reducing production incident detection time.
- Investigated and resolved production latency issues affecting SmartMatch and Contact Summarizer services across multiple AWS regions during Prime Day readiness.
- Led Amazon Security Review (ASR) compliance by driving Data Dictionary Review, remediating HTTPS and IAM findings, upgrading vulnerable dependencies, and strengthening production security.
- Migrated legacy recommender pipelines to rebuilt SageMaker endpoints while expanding SmartMatch availability through new marketplace launches and decommissioning legacy SageMaker V1 infrastructure.

Routing Resolution Models (Tier-1 ML Infrastructure)

- Led production-readiness initiatives to elevate Routing Models to Tier-1 infrastructure supporting mission-critical Customer Service AI traffic.
- Designed and integrated AWS Fault Injection Simulator (FIS) chaos tests into deployment pipelines, validating resilience through packet-loss and latency fault injection scenarios.
- Implemented automated rollback mechanisms and deployment safety improvements for public AWS deployment pipelines.
- Hardened deployment pipelines by improving approval workflows, resolving infrastructure automation issues, remediating OpenSSL vulnerabilities, upgrading dependencies, and strengthening platform security.
- Designed and implemented a configurable SageMaker Inference Recommender to optimize endpoint sizing, improving infrastructure utilization and deployment flexibility.

Human-in-the-Loop (HITL) Amazon Customer Service Chat Bot

- Root-caused a critical customer-impacting regression causing approved customer refunds to fail silently despite successful chatbot responses.
- Performed production analysis across nearly 1,000 customer sessions, identifying a multi-resolution model limitation and presenting findings to Applied Scientists for model remediation.
- Authored the technical design enabling multi-item HITL processing and resolved approval-token validation failures in the Concessions Orchestrator.
- Implemented eligibility rules for replacement, fully refunded, and zero-value orders while simplifying the HITL pipeline by retiring legacy experimentation logic after successful production rollout.

Catalog Experimentation and Impact Measurement Team (CEM)

Exakt Service Setup & Integration

- Led the end-to-end integration of Amazon's Exakt statistical engine into the Catalog Experimentation Platform, including architecture, implementation, and production deployment.
- Authored High-Level and Low-Level Design documents while provisioning infrastructure across beta and production environments using AWS CDK.
- Built Apache Airflow DAGs, Lambda workflows, and SNS integrations to automate experiment report generation and downstream processing.
- Developed CloudWatch dashboards, operational runbooks, and monitoring solutions improving production visibility and platform support.

- Validated statistical parity between Exakt reporting pipelines and baseline implementations through extensive production testing.
- Experimentation Platform Automation**
- Designed and implemented end-to-end experimentation workflows using EventBridge, AWS Lambda, Airflow, SNS, and S3 while automating Weblab creation, traffic dial-up, and dial-down.
 - Enhanced self-service experimentation capabilities through automated reporting, experiment lifecycle management, and production notification workflows.
 - Improved workflow reliability through Step Functions redesign, expanded integration testing, and operational monitoring.
 - Led implementation of No_Value support for Catalog V3 completeness experiments and executed production validation across complex catalog attribute types.
 - Resolved production issues involving DynamoDB consistency, Step Functions execution, Airflow scheduling, SQS visibility, experiment synchronization, CloudAuth migration, IAM remediation, and platform hardening while authoring deployment documentation and operational runbooks.
- Amazon Scout Robotics**
- Teleoperations & Delivery Experience Team**
- Developed integration tests supporting teleoperation and delivery workflows for Amazon Scout robotics.
 - Implemented Java and TypeScript features for robotics management systems while collaborating with cross-functional engineering teams.

Projects

AI Support Inbox | Full-Stack AI Customer Support Platform

Tech Stack: Java 17, Spring Boot, PostgreSQL, React, TypeScript, Ollama (Llama 3), REST APIs

- Developed a full-stack AI-powered customer support application that enables support agents to manage tickets and generate context-aware AI responses.
- Built a RESTful backend using Spring Boot and PostgreSQL to support ticket creation, retrieval, status tracking, and priority management.
- Integrated a locally hosted Llama 3 large language model through Ollama to generate intelligent draft responses while implementing robust error handling and graceful fallback mechanisms.
- Designed a responsive frontend using React and TypeScript featuring an interactive support inbox, ticket details, and one-click AI-assisted reply generation.
- Implemented clean API design and end-to-end frontend/backend integration following modern full-stack development practices.

Technologies: Java, Spring Boot, PostgreSQL, React, TypeScript, REST APIs, Ollama, LLM

Links to Images : [\[Chat View\]](#)

Life Calendar | Serverless Personal Milestone Tracking Application

Tech Stack: Python, FastAPI, AWS Lambda, DynamoDB, React, JWT, Google OAuth, Vercel

- Developed a full-stack serverless application that visualizes an individual's life as weekly milestones, enabling users to capture and reflect on significant life events.
- Designed an intuitive calendar-based interface that organizes milestones by age and month while efficiently supporting large milestone datasets.
- Built a FastAPI backend deployed on AWS Lambda with DynamoDB as the persistence layer for scalable, serverless data storage.
- Implemented secure authentication using Google OAuth and JWT-based authorization to protect user data and application endpoints.
- Managed CORS, authentication flows, and API security while deploying the frontend on Vercel and backend using AWS serverless infrastructure.

Technologies: Python, FastAPI, AWS Lambda, DynamoDB, React, Google OAuth, JWT, Vercel

Links to Images : [\[Login View\]](#) , [\[Calendar View\]](#) , [\[Add Milestone View\]](#) , [\[Milestone List View\]](#)

Skills	
Programming Languages	Backend Development
Java, Python, Typescript, R, SQL, Javascript, Datapth query language	Spring Boot, Spring, REST APIs, FAST API, Microservices, Docker
AWS & Cloud	AI / Machine Learning
Amazon SageMaker AI, AWS Lambda, S3, CloudFormation, AWS CDK, Cloudwatch, EventBridge, DynamoDB, SNS, SQS, StepFunctions, Apache Airflow, IAM, Glue, Athena, RDS, EC2, EMR	Large Language Models (LLMs), Amazon SageMaker AI, Prompt Engineering, Ollama, Mistral, Inference Pipelines, Model Deployment, PreProcessing, PostProcessing
Databases & Data Engineering	DevOps & Tools
PostgreSQL, DynamoDB, Apache Airflow, EMR, Apache Spark	Git, GitHub, CI/CD pipelines, Docker, Linux, Cloudwatch, JUnit, Mockito

Education	
Northern Illinois University	M.S
Computer Science	DeKalb, IL • 2021-2022
Institute of Aeronautical Engineering	B.Tech
Computer Science	Hyderabad, India • 2016-2020